WILLIAM ELIAS

SOFTWARE & AUTOMATION ENGINEER

Developer Tooling | *Internal Applications* | *Platform Engineering* | *Python* | *Go* | *C#/.NET*
Michigan | Open to U.S. Remote Opportunities | WylElias.123@gmail.com
linkedin.com/in/wylelias | github.com/howlcipher | howlcipher.github.io/william_elias

PROFESSIONAL SUMMARY

Software and automation engineer building developer tools, internal applications, and automation, with software and automation work since 2020 and production and infrastructure experience dating to 2015. Builds a Go self-service deployment CLI, Python/FastAPI dashboards and Azure DevOps REST automation, and a C#/.NET Blazor internal application with access policies and audit history. Turns CI/CD standardization, credential remediation, and recurring manual work into reusable pipeline templates, CLI tooling, and scheduled automation. Adds security depth through credential-remediation tooling, a scoped OAuth 2.1 resource server with JWT/JWKS validation, and in-progress M.S. Cyber Defense studies.

PROFESSIONAL EXPERIENCE

Stellantis Financial Services US Feb 2023 - Present

Senior Production Support Engineer | *DevOps & Automation* | *Auburn Hills, MI · Hybrid*

Promoted September 2026; previously Production Support Engineer

- Designed and delivered a Go-based self-service deployment CLI covering the build-to-rollback lifecycle with guided setup, Azure DevOps REST integration, config parsing, pipeline generation, build/push/deploy, rollback, status, audit, and post-deployment validation.
- Built and maintain an internal .NET 8 Blazor Web App with Interactive Server, C#, ASP.NET Core, SQL Server, and SQLite for production support, with per-module access policies, database-backed workflows, retained audit history, live configuration/object diffing, pre-change/pre-CAB visibility, and service-desk request tracking.
- Built Python/FastAPI dashboards and operational tooling separate from the Blazor portal, spanning Azure DevOps REST API automation, failed-payload triage, scheduled query-to-email reporting, XML integration-log parsing, ranked engineering search, automated RCA, reusable operations CLI functionality, and recurring production-support workflows; standardized the team on uv for Python dependency and tooling management.
- Built Python tooling to scan source code and Git history for exposed credentials and used BFG Repo-Cleaner to scrub historical records across 100+ repositories. A measured scan processed 2,832 files and identified 67 distinct exposed secrets in approximately 25 seconds; remediation records are complete.
- Designed a scoped OAuth 2.1 resource server with JWT/JWKS validation, content allowlisting, and explicit token-validation-bypass defenses.
- Directed a CI/CD standardization program across a 60-repository, 104-application internal estate, building reusable pipeline/template automation and 56 Azure DevOps build/release definitions across 28 standardized application repositories.
- Executed and validated builds across all 28 repositories (27 passed, with the remaining failure diagnosed and logged); dry-run validated representative deployment paths for 25 of 28, with the remaining checks blocked by upstream/environment or tooling conditions. Reviewed and classified 109 existing definitions, identifying 95 legacy and 14 already-aligned definitions, and resolved several latent delivery defects spanning path-handling assumptions, configuration drift, and false-success states.
- Maintain the release-tracking process for a recurring, roughly weekly production release cadence and participate directly in production deployments. Authored and maintain a reusable 22-query KQL library used across internal monitoring and troubleshooting automation.

HBK - Hottinger Brüel & Kjær Jul 2020 - Feb 2023

DevOps Engineer - Python Automation | *Southfield, MI · Remote*

- Created Python/Tkinter XML-output automation that reduced data-processing time by 60%; automated SQL output processing to save more than 40 hours per month.
- Developed database-configuration tools to streamline customer onboarding and reduce repetitive environment setup.
- Built NSIS installer/uninstaller packages, maintained Git workflows and developer tooling, and provided live customer troubleshooting.

CORE EXPERTISE

Software & Backend: Python, Go, C#, .NET, ASP.NET Core, Blazor, FastAPI, REST APIs, SQL Server, SQLite, JavaScript, HTML, CSS

Automation & Developer Tooling *(Python/PowerShell professionally; Go for deployment tooling; uv for Python standardization)***:** Python, PowerShell, REST APIs, Go, uv, Developer Tooling
DevOps & Delivery *(Azure DevOps professionally; containerization proof of concept)***:** Azure DevOps, CI/CD, Release Automation, YAML Pipelines, Git, Containerization
Security & Identity: OAuth / OIDC, IAM, Credential Remediation, Git History Remediation, PII Safeguards, Audit Controls
AI-Enabled Engineering *(Internal tools & projects)***:** LLM APIs, MCP, RAG, Agent Workflows, Human Review, Engineering Knowledge Systems
Production & Reliability: Azure Monitor, Application Insights, KQL, Structured Logging, Incident Triage, RCA

SELECTED ENGINEERING HIGHLIGHTS

Self-Service Deployment CLI (Go)

- Guided setup and project initialization/scaffolding generate pipeline configuration; Azure DevOps REST integration drives build, push, deploy, rollback, status, and audit; host auditing, fail-fast deployment-path validation, container-lifecycle integration, and automated post-deployment verification are built in.

Tech: Go, Azure DevOps, REST APIs, CI/CD, IIS

Internal Web Application with Access Policies & Audit (C#/.NET)

- Change-safety features in the modular .NET 8 Blazor Web App include DBA approve-only deployment scripts, live configuration/object diffing for pre-CAB and pre-deployment review, per-module access policies, and retained audit history; integrated service-desk request tracking keeps operational workflows traceable.

Tech: C#, .NET 8, Blazor, ASP.NET Core, SQL Server, SQLite

Python Developer Productivity Tooling

- Sole-authored Python tools parse failed integration payloads into a structured DBA-approval queue, run scheduled query-to-email reports, and summarize processing errors embedded in XML integration logs; repository-discovery and Git-history remediation tooling supports single-repository and whole-project modes with interchangeable rewrite engines.

Tech: Python, REST APIs, Azure DevOps, Automation, uv

SELECTED OPEN-SOURCE ENGINEERING

HowlPlane

github.com/howlcipher/howlplane

- Built a Python/Go engineering control plane that routes work across coding agents, reconciles independent reviews, and runs deterministic verification.
- Added durable evidence records, human approval gates for consequential actions, and CI evidence bound to the pull-request head commit.

HowlFrame

github.com/howlcipher/howlframe

- Built an experimental language and compiler toolchain in Go with a lexer, parser, typed AST/IR, bytecode VM, WebAssembly target, and a capability-based security model enforced by the VM.
- Explores constrained AI code generation against typed IR, with local Ollama integration.

EARLIER EXPERIENCE

QA Developer, Intrepid Control Systems Mar 2020 - Apr 2020

- Wrote Mocha tests and debugged APIs across software and hardware integrations.

Network Engineer, Project Worldwide Jan 2020 - Mar 2020

- Configured Cisco/Meraki infrastructure and executed multi-state network migrations with minimal downtime.

Network Standards Engineer, Ford Motor Company Sep 2019 - Dec 2019

- Implemented global network standards, documentation, and IP-management systems.

IT Network Engineer, Trendset Communications Group Sep 2015 - Jun 2019

- Maintained network standards, configurations, and documentation; supported critical production infrastructure.

EDUCATION & CERTIFICATIONS

M.S. Cyber Defense (In Progress, Second Year) - Dakota State University (Expected Winter 2027)
B.S. Information Technology - Colorado State University Global Campus
B.B.A. Business Administration - Rochester College
CCNA (Previously Held) - Cisco Networking Academy (2014 - 2017)